

Approximate Linearization Control for Autonomous Hockey with a RoboMaster EP Mobile Manipulator

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Abstract—This project develops and implements an approximate linearization controller for a RoboMaster EP mobile manipulator to complete an autonomous hockey task. The robot is required to navigate to the hockey stick base, grab the stick, navigate to the puck, and shoot the puck into a goal. To complete the task, the same approximate linearization controller is used for all navigation stages, and additional geometric planning methods are developed to generate the target positions for stick grabbing and puck shooting. An arm and gripper sequence is also designed for grabbing the stick and executing the shooting motion. The controller is validated both in the simulator and on the physical RoboMaster EP in the RobotHub. Experimental results show that the robot is able to complete the entire hockey task autonomously after calibrating some parameters.

I. PROPOSED APPROACH

A. System Overview

The purpose of this project is to enable a RoboMaster EP mobile manipulator to autonomously complete a hockey task. The robot is required to navigate to the hockey stick and grab it, then navigate to the puck, and shot the puck into the goal. The poses of the robot, stick base, puck, and the goal are known and obtained from the Vicon motion-capture system. The controller is implemented in ROS2 environment and sends commands to the robot.

The proposed controller divides the task into two main stages: stick grabbing and puck shooting. During the grabbing stage, the robot first opens the gripper and raises the arm. Then, it navigates to a pre-grabbing position located along the outward normal of the stick base. Once it reaches this point, the robot will rotate in place to align with the orientation of the base and make a slower final approach. After the required clearance is reached, the arm moves forward and downward, closes the gripper to grab the stick, and then the arm moves backward and upward to lift it. The robot then moves backward to the pre-grabbing position. The purpose of this action is to create sufficient clearance from the stick base and prevent the stick from hitting the stick base when the robot align with the next navigation target in next stage.

During the shooting stage, the controller uses the puck and goal positions to determine a desired shooting direction. Based on this direction, it calculates a pre-shooting position and a final-shooting position on one side of the puck that perpendicular to the shooting direction. The robot navigates to the pre-shooting position, aligns the raised stick toward

the puck, and performs a slow 180-degrees rotation. Then, it moves backward toward the final-shooting position, lowers the stick to the floor, and performs a fast 270-degree rotation to strike the puck into the goal.

The controller is implemented using a finite state machine. The robot completes the task step by step, including navigation, grabbing the stick, and shooting the puck. The controller will only switch to the next stage after the previous stage is fully completed, until the entire task is completed. If any stage cannot be completed for any reason, such as environment failure, the controller will stop the entire task and output an error message, to ensure the continuity and safety of the experiment.

B. Approximate Linearization-Based Navigation

Navigation between different positions is achieved by the approximate linearization controller. This controller is used whenever the robot needs to move to a desired position, including the stick pre-grabbing position, the grabbing position, the pre-shooting position, or the final-shooting position.

The RoboMaster EP is modeled as a unicycle with the model

$$\begin{cases} \dot{x} = v \cos \theta \\ \dot{y} = v \sin \theta, \\ \dot{\theta} = \omega \end{cases}$$

where x and y represent the position, and θ represent the orientation of the robot, and v and ω are the linear and angular velocity inputs respectively.

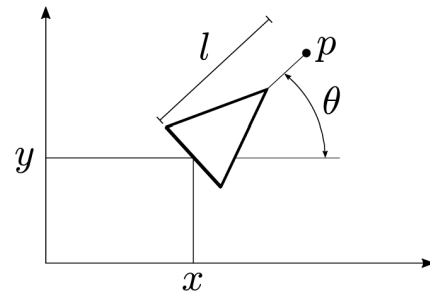


Fig. 1. Geometric parameters of the robot required to derive the approximate linearization controller.

Following the approximate linearization controller, we define a virtual control point in front of the robot

$$\mathbf{p} = \begin{bmatrix} x_l \\ y_l \end{bmatrix} = \begin{bmatrix} x + l \cos \theta \\ y + l \sin \theta \end{bmatrix},$$

where l is the distance between the robot and the virtual control point. Instead of directly controlling the robot itself, the controller drives this virtual control point toward the desired target. Differentiating the virtual control point gives

$$\dot{\mathbf{p}} = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & l \end{bmatrix} \begin{bmatrix} v \\ \omega \end{bmatrix} = R(\theta)L(l) \begin{bmatrix} v \\ \omega \end{bmatrix},$$

or equivalently

$$\begin{bmatrix} v \\ \omega \end{bmatrix} = L(l)^{-1}R(\theta)^\top \dot{\mathbf{p}}.$$

The desired velocity of the virtual control point is generated using a proportional controller

$$\dot{\mathbf{p}} = K_p(\mathbf{p}_d - \mathbf{p}),$$

where $\mathbf{p}_d$ is the desired target position and K_p is the proportional gain.

To sum up, the controller continuously computes the linear and angular velocity inputs that drives the robot toward the desired target position. In our implementation, the same controller is used throughout the task, while only the target position changes at the different stage of the task. During the grabbing stage, the controller is used to navigate the robot to the pre-grabbing position and the grabbing position. During the shooting stage, it is used to navigate to the pre-shooting position and the final-shooting position. Different velocity limits and stopping tolerances are also used for different stages for accuracy and safety concerns.

C. Stick Grabbing Planning and Execution

The stick grabbing stage includes two navigation procedures and a predefined arm and gripper sequence. Because the approaching orientation is important for the robot to successfully grab the stick, instead of navigating directly toward the stick base, the robot first navigates to a pre-grabbing position. This gives the robot enough space to adjust its orientation which ensures that the robot approaches the stick base from orthogonal direction.

Let the position of the stick base be $\mathbf{p}_s = [x_s \ y_s]^\top$, and θ_s be its orientation. Based on this orientation, the outward unit orthogonal direction to the stick base is

$$\mathbf{n}_s = \begin{bmatrix} \cos \theta_s \\ \sin \theta_s \end{bmatrix}.$$

To compensate for the position error that exists in real experiments, the position of the stick base $\mathbf{p}_s$ is shifted slightly along the tangential direction, and use the shifted position to generate the navigation targets,

$$\tilde{\mathbf{p}}_s = \mathbf{p}_s + \delta_s \mathbf{t}_s$$

where δ_s is a small offset determined in the real experiment, and

$$\mathbf{t}_s = \begin{bmatrix} -\sin \theta_s \\ \cos \theta_s \end{bmatrix}.$$

is the unit tangential direction that orthogonal to $\mathbf{n}_s$. The pre-grabbing position is then calculated by

$$\hat{\mathbf{p}}_{\text{grab}} = \tilde{\mathbf{p}}_s + \hat{d}_{\text{grab}} \mathbf{n}_s,$$

where $\hat{d}_{\text{grab}}$ is the distance from stick base to the pre-grabbing position, which requires to be measured experimentally.

After reaching the pre-grabbing position $\hat{\mathbf{p}}_{\text{grab}}$, the robot rotates in place to face to the center of the adjusted stick base $\tilde{\mathbf{p}}_s$, and then performs a slower final approach toward the grabbing position which defined as

$$\mathbf{p}_{\text{grab}} = \tilde{\mathbf{p}}_s + d_{\text{grab}} \mathbf{n}_s,$$

where $d_{\text{grab}} < \hat{d}_{\text{grab}}$ is the distance from stick base that robot stops and grab the stick. The final grabbing distance is selected so that the robot stops close enough to able to grab the stick while avoid hit the base. The final approach uses a lower linear velocity than the previous navigation stage. The robot also continuously corrects its heading toward the center of the base during this motion.

Once the robot reaches the grabbing position $\mathbf{p}_{\text{grab}}$, it performs the predefined arm and gripper sequence to grab the stick. Since the gripper is first opened and the arm is raised before navigation start to avoid collision with stick base. At the grabbing position, the arm moves diagonally forward and downward so that the gripper can approach the stick. Next, the gripper closes and grabs the stick, and then the arm moves diagonally backward and upward to lift the stick from the base.

After the stick is lifted, the robot moves backward along the same approach direction until it returns to the original pre-grabbing position $\hat{\mathbf{p}}_{\text{grab}}$. This will provide sufficient clearance between the robot and the stick base before the shooting stage begins, to prevent the stick from colliding with the base due to lateral sweeping.

D. Shot Geometry and Target Generation

After the stick is grabbed, the controller will first calculate where to navigate based on the positions of the puck and the goal. Instead of directly navigate to the puck, the controller first computes the desired shooting direction and then generates two navigation targets that prepare the robot for the final shooting motion.

Let the position of the puck and goal be $\mathbf{p}_p = [x_p \ y_p]^\top$ and $\mathbf{p}_g = [x_g \ y_g]^\top$ respectively, then the desired unit shooting direction can be calculated as

$$\mathbf{r}_p = \begin{bmatrix} r_{x_p} \\ r_{y_p} \end{bmatrix} = \frac{\mathbf{p}_g - \mathbf{p}_p}{\|\mathbf{p}_g - \mathbf{p}_p\|},$$

and the unit direction that orthogonal to the shooting direction is

$$\mathbf{n}_p = \begin{bmatrix} r_{y_p} \\ -r_{x_p} \end{bmatrix}.$$

The robot was designed to approach the puck from this orthogonal direction, so that the puck will be shot through the shooting direction $\mathbf{r}$ when the stick rotate and hit the puck.

To compensate for the position error observed during real experiments, the puck position is shifted slightly along the shooting direction before generating the navigation targets

$$\tilde{\mathbf{p}}_p = \mathbf{p}_p + \delta_p \mathbf{r}_p,$$

where δ is a small offset determined in the real experiment.

Based on the adjusted puck position, two target positions are generated. The first one is the pre-shooting position,

$$\hat{\mathbf{p}}_{\text{shoot}} = \tilde{\mathbf{p}}_p + \hat{d}_{\text{shoot}} \mathbf{n}_p,$$

which allows the robot to align itself before the shooting motion. The second one is the final shooting position,

$$\mathbf{p}_{\text{shoot}} = \tilde{\mathbf{p}}_p + d_{\text{shoot}} \mathbf{n}_p,$$

where $d_{\text{shoot}} < \hat{d}_{\text{shoot}}$. The robot first navigates to the pre-shooting position and then moves backward to the final shooting position before lowering the stick to the ground.

By separating the navigation into two stages, the robot can first achieve a stable heading and then accurately position the center of rotation for the final swing. This improves the repeatability of the shooting motion and produces a more consistent puck trajectory toward the goal.

E. Shooting Sequence

After reaching the pre-shooting position $\hat{\mathbf{p}}_{\text{shoot}}$, the robot first calibrates its orientation by aligns its heading so that the raised stick points towards the adjusted puck position $\tilde{\mathbf{p}}_p$. This orientation is used as the reference orientation of the wind-up rotation which operates next and also ensures that the stick follows the expected trajectory during rotation.

After calibration, the robot performs a slow 180-degree wind-up rotation. After the wind-up rotation, the robot move reversely toward the final shooting position $\mathbf{p}_{\text{shoot}}$. The reason for moving reverse instead of moving forward is to prevent the stick from accidentally hitting the puck before the final shot. Once the robot reaches the final shooting position $\mathbf{p}_{\text{shoot}}$, the stick is lowered to the ground and pauses briefly before the final spin action. This ensures that the stick is in full contact with the ground at the moment of hitting the puck.

Finally, the robot performs a fast 270-degree rotation to strike the puck. During this motion, the stick sweeps through the puck and transfers momentum in the shooting direction $\mathbf{r}_p$. The reason for rotating 270 degrees rather than 180 degrees is to ensure that the stick is able to hit the puck without speed loss.

The two rotation actions described above will stop once the measured angle reaches the expected value, instead of depending on an open-loop timing command.

II. RESULTS

A. Stick Grabbing Stage

The complete stick grabbing stage was tested in the simulator for multiple times, with different initial robot locations and stick locations. In all tests, the robot successfully navigated to the pre-grabbing position, aligned the orientation, reached

the grabbing position, and executed the predefined arm and gripper sequences. These experiments illustrate that the current controller is able to complete the stick grabbing stage under different initial conditions.

The proposed controller was also validated on the physical RoboMaster EP mobile manipulator in the RobotHub session. Several additional measurements and calibrations were needed for the physical robot. The parameters that used in the stick grabbing stage, including the pre-grabbing distance $\hat{d}_{\text{grab}}$ and grabbing distance d_{grab} , and the horizontal offset δ_s were measured manually. After calibrate these parameters, the robot was able to navigate to the desired grabbing position, align the gripper with the stick, and successfully grab and lift the stick in the real environment. These results demonstrate that the proposed stick grabbing stage can be effectively used to the physical robot with some calibration.

B. Puck Shooting Stage

The proposed puck shooting stage was then validated in the simulator after the robot completed the stick grabbing stage with different puck and goal locations. During the tests, the robot successfully navigated to the pre-shooting position $\hat{\mathbf{p}}_{\text{shoot}}$, calibrated orientation, performed the wind-up rotation, moved backward to the final shooting position $\mathbf{p}_{\text{shoot}}$, and executed the final shooting motion. The puck moved along the desired shooting direction toward the goal, which proved that the controller provides a reliable shooting stage under different tests.

The proposed shooting strategy was also evaluated on the physical RoboMaster EP mobile manipulator in the RobotHub. Similar to the stick grabbing stage, the parameters $\hat{d}_{\text{shoot}}$, d_{shoot} , and a offset δ_p for the puck position were experimentally measured and tuned to calibrate the real robot. After that, the robot was able to reach the desired shooting position and complete the predefined shooting sequence. During the real experiments, the puck did struck in the intended direction toward the goal. However, the RoboMaster EP has a maximum rotational speed limit, which sometimes causes the impact velocity to not sufficient for the puck to reach the goal. But, the results indicate that the proposed puck shooting stage generates the correct shooting direction. If the robot can reach a higher allowable rotation speed, the same controller is expected to shoot the puck into the goal.

C. Complete Task Execution

After validating the stick grabbing stage and the puck shooting stage separately, the two controllers were combined into a single finite-state controller to perform the complete hockey task. Following figures shows a representative execution of the combined controller, including trajectory, robot pose, and controller input values.

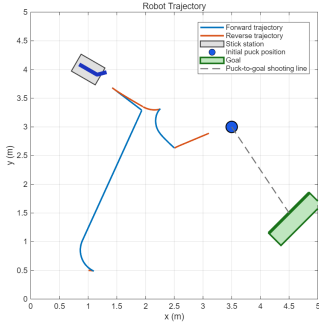


Fig. 2. Robot trajectory with initial conditions $[x \ y \ \theta] = [1 \ \frac{1}{2} \ \pi]$, $[\mathbf{p}_s^\top \ \theta_s] = [1 \ 4 - \frac{\pi}{6}]$, $\mathbf{p}_p^\top = [\frac{7}{2} \ 3]$, $[\mathbf{p}_g^\top \ \theta_g] = [\frac{9}{2} \ \frac{3}{2} - \frac{\pi}{4}]$.

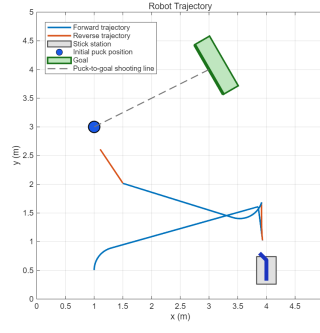


Fig. 3. Robot trajectory with initial conditions $[x \ y \ \theta] = [1 \ \frac{1}{2} \ \frac{\pi}{2}]$, $[\mathbf{p}_s^\top \ \theta_s] = [4 \ \frac{1}{2} \ \frac{\pi}{2}]$, $\mathbf{p}_p^\top = [1 \ 3]$, $[\mathbf{p}_g^\top \ \theta_g] = [3 \ 4 \ \frac{\pi}{4}]$.

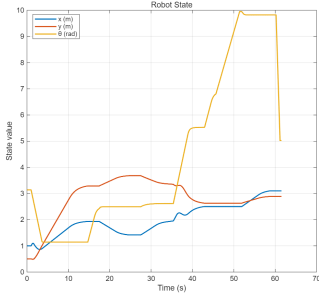


Fig. 4. Robot state with initial conditions $[x \ y \ \theta] = [1 \ \frac{1}{2} \ \pi]$, $[\mathbf{p}_s^\top \ \theta_s] = [1 \ 4 - \frac{\pi}{6}]$, $\mathbf{p}_p^\top = [\frac{7}{2} \ 3]$, $[\mathbf{p}_g^\top \ \theta_g] = [\frac{9}{2} \ \frac{3}{2} - \frac{\pi}{4}]$.

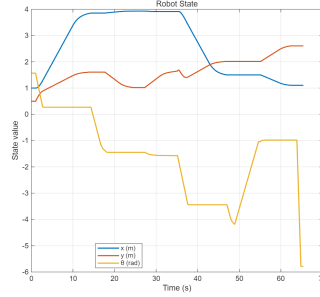


Fig. 5. Robot state with initial conditions $[x \ y \ \theta] = [1 \ \frac{1}{2} \ \frac{\pi}{2}]$, $[\mathbf{p}_s^\top \ \theta_s] = [4 \ \frac{1}{2} \ \frac{\pi}{2}]$, $\mathbf{p}_p^\top = [1 \ 3]$, $[\mathbf{p}_g^\top \ \theta_g] = [3 \ 4 \ \frac{\pi}{4}]$.

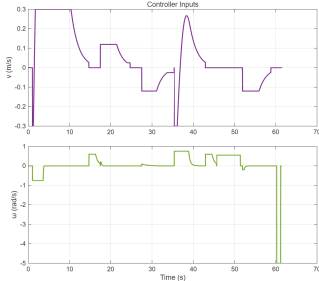


Fig. 6. Control input with initial conditions $[x \ y \ \theta] = [1 \ \frac{1}{2} \ \pi]$, $[\mathbf{p}_s^\top \ \theta_s] = [1 \ 4 - \frac{\pi}{6}]$, $\mathbf{p}_p^\top = [\frac{7}{2} \ 3]$, $[\mathbf{p}_g^\top \ \theta_g] = [\frac{9}{2} \ \frac{3}{2} - \frac{\pi}{4}]$.

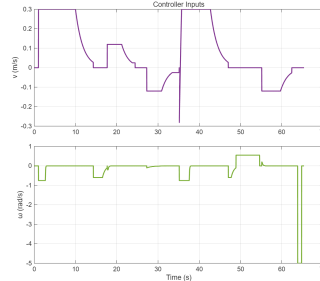


Fig. 7. Control input with initial conditions $[x \ y \ \theta] = [1 \ \frac{1}{2} \ \frac{\pi}{2}]$, $[\mathbf{p}_s^\top \ \theta_s] = [4 \ \frac{1}{2} \ \frac{\pi}{2}]$, $\mathbf{p}_p^\top = [1 \ 3]$, $[\mathbf{p}_g^\top \ \theta_g] = [3 \ 4 \ \frac{\pi}{4}]$.

Starting from initial position, the robot successfully navigated to the stick, grab it, reversed to the pre-grabbing position, navigated to the puck, and shoot the puck to the goal. All actions are completed automatically, without any manual operation. These experiments shows that the combined controller is able to complete the entire hockey task using a unified control framework.

III. DISCUSSION

A. Advantages

The proposed controller adopts the control framework based on approximate linearization. The same controller is used

throughout the task, and all different behaviors in different navigation process are based on changing the target position. This simplifies the overall controller design. In addition, the controller is organized as a finite-state machine that divides the entire hockey task into stick grabbing stage and puck shooting stage, and each stage also has its subdivisions. This structure simplifies the implement and debug process, and allows each stage to be validated independently before combining them together. Finally, the proposed controller was successfully validated in both the simulator and the physical RoboMaster EP mobile manipulator. The experimental results show that the controller can be applied from simulation to the real robot with only some parameter calibration.

B. Limitations

One limitation of the proposed controller is that it assumes that the puck is already in a position where it can be shot directly toward the goal, or the shooting angle is large enough. However, if the shooting angle is too small, or negative, the current strategy may not be suitable.

Another limitation is that obstacle avoidance is not considered in the current controller. During our experiments, several robots operated in the workspace at the same time. Sometimes, they collide with each other when their paths cross which hugely interrupted the validation experiments.

C. Future Work

Some future improvements can be considered on the current controller. For example, improving the shooting strategy for more complex puck positions or adding obstacle avoidance during navigation.

A possible improvement to the shooting strategy is to add a shooting angle calculation process. Based on the shooting direction $\mathbf{r}_s$ and the angle of the goal θ_g , the shooting angle can be calculated. If the angle is too small, then the controller should set a target point which with a larger shooting angle, and move the puck to that location before the final shot.

Another possible improvement is to add obstacle avoidance to the controller. One way to achieve this is to introduce a potential function for the workspace. The target position is assigned a low potential value, and obstacles, such as other robots, are assigned high potential values. An available potential value can be the reciprocal of the distance to the obstacle. During navigation, instead of moving directly toward the target, the controller navigates the robot in the direction with lower potential values. Therefore, when other robots are on the planned path, our robot will move around it instead of following a fixed path. Since the locations of all robots are available from the Vicon system, the potential field can be updated continuously during the experiment.

IV. AUTHOR CONTRIBUTIONS

All the work including controller implementation, simulator development, experiments, data collection, and report writing were completed by Hanhua Ge since the other group member withdrew from the project, and the instructor is aware of this matter.